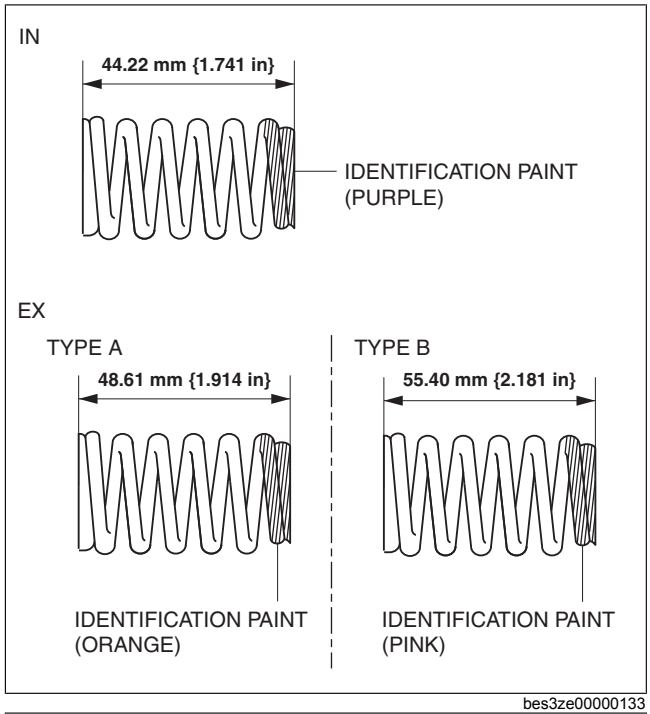
VALVE SPRING INSPECTION

id011000501400

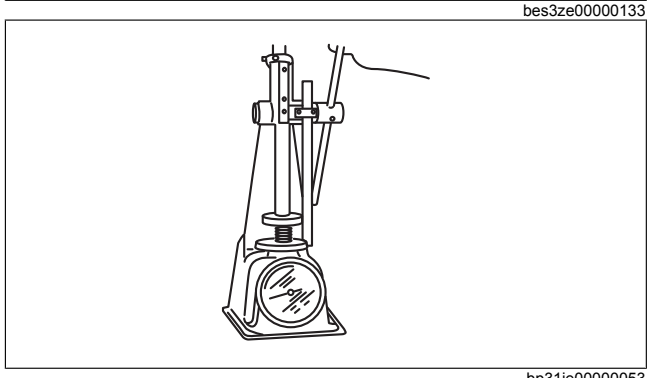
Caution
- **The valve springs differ depending on the IN and EX sides. Therefore, verify the free length or identification paint beforehand and inspect the valve springs.**

1. Measure the valve spring height using the spring tester.
   - If it is not within the specification, replace the valve spring.

   Valve spring installation height
   IN: When pressurized with spring force of 128.3—141.7 N {13.09—14.44 kgf, 28.85—31.85 lbf}, spring height is 35.0 mm {1.38 in}
   EX (TYPE A): When pressurized with spring force of 137.8—152.2 N {14.06—15.52 kgf, 30.98—34.21 lbf}, spring height is 35.0 mm {1.38 in}
   EX (TYPE B): When pressurized with spring force of 209—231 N {21.4—23.5 kgf, 47.0—51.9 lbf}, spring height is 35.0 mm {1.38 in}



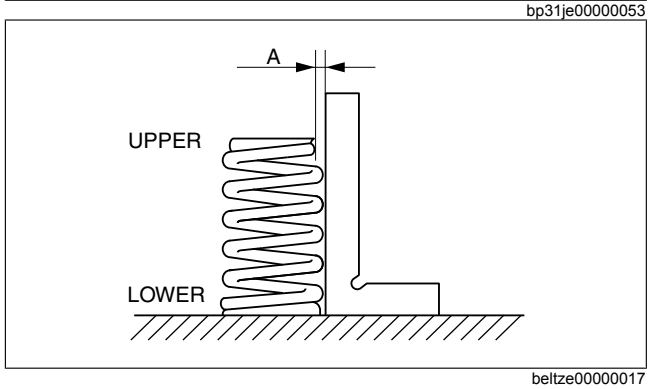
bes3ze00000133



bp31je00000053

2. Measure the amount of off-square on the valve spring using a square.
   (1) Rotate the valve spring one full turn and measure A at the point where the gap is the largest.
      - If it exceeds the maximum specification, replace the valve spring.

   Maximum valve spring off-square
   IN: 2.0 ° (1.5 mm {0.059 in})
   EX (TYPE A): 2.0 ° (1.7 mm {0.067 in})
   EX (TYPE B): 2.0 ° (1.9 mm {0.075 in})



beltze00000017